

WHERE THE LOOPS COME FROM

Why some networks branch and others join back up

THE PUZZLE

LOOK at frost on a cold window. It grows outwards in arms that split again and again, and the arms usually do not join back up. Wherever a tip pokes slightly ahead of its neighbours, it finds more water vapour, grows faster and pulls further ahead. The gaps between the arms are left starved. That is the whole rule, and it is a local one. Nothing is planning the pattern.

Frost makes a branching shape, but here is the strange thing about it: nothing travels along a frost arm. It is not a road or a pipe. It is just a record of where growing happened to be easy. Every other network in this article is different, because every other one has a job to do. Something has to get from one place to another along it.

A rule about loops

Draw a network of any kind and ask one question: can you follow different links away from a point and return without turning back? If you cannot, the network is a **tree**: all branches and no rings. If you can, it has a **loop**.

Loops cost something. Every extra connection needs material to build and maintain. If your only aim were to use as little as possible, you would not build one. This can be proved: in any loop, at least one link can be deleted and all the points will still be connected.

So why does anything ever build a loop? Hold that question. It is the central question in this article.

Rivers: almost no loops

Rivers branch beautifully. Small streams join into bigger streams all the way towards the sea, but an ordinary river system almost never contains a ring. Braided channels, flood plains and deltas are important exceptions. Away from them, the pattern is very close to a tree.

The reason is that a river has a fairly predictable transport problem. Water mainly goes downhill, so the overall direction does not reverse. The main outlet stays in the same place, and tomorrow's job is much like today's. A spare route rarely helps. When a network only needs to carry water one way towards a stable outlet, a tree is usually the cheapest pattern.

Slime mould: loops, but not many

A slime mould called *Physarum* is one large cell containing many nuclei. It spreads across a forest floor, or across a Petri dish in a laboratory. Put oat flakes down in scattered places and it has to reach all of them, without a brain, without eyes and without knowing where the next flake will appear.

Its method is almost embarrassingly simple. First it spreads everywhere, sending tubes in all directions. Then it thickens any tube that carries a lot and lets the quiet ones wither away. What remains is short and efficient, but it keeps a few extra connections that a shortest-route calculation would not have paid for.

Japanese researchers once laid oat flakes on a dish in the pattern of towns around Tokyo. The network the mould grew was similar to the Tokyo rail network in three measurable ways: total length, how quickly material could move between points, and how well the network survived having a link cut.

The mould was not being clever, and it was not designing a railway. It was following a simple rule. That rule happened to produce a good answer to a problem rather like the one faced by railway engineers.

The rule behind the rule

Researchers have worked out when loops can be worth their cost. The answer is short.

If the traffic stays the same and every route can be trusted, a tree is often the cheapest network.

If the traffic changes, or if a link might break, loops can be worth building.

That is the idea. Loops are what you build when you do not know exactly what will be asked of the network. A leaf's veins contain loops for this reason. So do many of the blood-vessel networks in your body, and so does the slime mould.

Under the forest, with a warning

Fungi grow enormous underground networks, often linked with plant roots. They can behave rather like the slime mould, only over metres and years instead of a dish and a weekend. They strengthen useful routes, abandon others and keep some spare connections. A fungus cannot know where the next dead leaf will land or where an animal will dig.

You may have heard this called the "wood-wide web", with older "mother trees" deliberately feeding nutrients to seedlings and sending warnings to their neighbours. Be careful here. Scientists reviewing the best-known studies found that the popular story is much stronger than the evidence.

What is clearly true is that fungal networks exist, can connect roots and move carbon underground. What remains doubtful is how much moves directly from tree to tree, in which direction, who benefits and whether any of it is deliberate.

The safest simple reading is that the fungus is running its own business, and the trees are customers rather than family. The loops still make sense: insurance is useful whether you are generous or selfish.

The one sentence

A loop is an admission that the future is unknown.

That is why loops appear where something must be delivered but nobody can say in advance where it will be needed, and why ordinary river networks usually do without them.

Cities: everybody solves a different piece

City streets are full of loops, and by now you can say why. People are going to different places, destinations change by the hour, everyone can be both a starting point and an ending point, and any road can be closed.

A city is different from the other examples in one important way: nobody solves the whole problem. Cities do have planners, but no single plan covers their entire history. Each generation solves smaller local problems while working around river crossings, old field edges and walls that may no longer exist. That is why one city can contain a Roman grid, a medieval tangle and a modern ring road within a short walk.

Cities differ, but researchers still find repeated patterns in their road networks and in the amount of infrastructure they need. No single rule fits every city. The useful point is that many local decisions can still produce a large-scale pattern.

What it all comes down to

Four systems, from chemistry to geography, and only two questions separate them.

Does anything travel along it? Frost, no. Everything else here, yes. Frost's branching answers only to the water vapour and temperature around it.

Is the traffic predictable? For a river, it is fairly predictable, so a tree is usually enough. For a slime mould, a fungus or a city, it is not. Spare routes can matter, so loops appear.

What differs between those last three is the mechanism and the timescale: hours and centimetres for the mould, seasons and metres for the fungus, centuries and square kilometres for the city. The same broad answer is reached by a cell with no brain, a fungus with no central planner, and a crowd of people none of whom is in charge.

SELECTED EVIDENCE

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